

MH1100 Calculus I

Handout 11 - Application of Differentiation

Application of Differentiation

- Optimization Problems
- Newton's Method
- Antiderivatives

Application

Further readings:

**Mathematical
optimization;
Linear programming;
Convex optimization;**



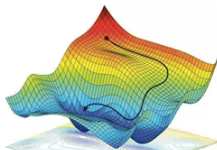
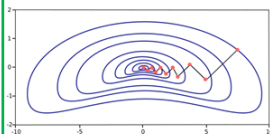
Logistics



❖ **Optimization Problems**

Further readings:

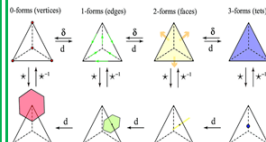
**Numerical analysis;
Finite difference method;
Finite element methods;
Gradient descent;**



❖ **Newton's Method**

Further readings:

**Integration;
Riemann integral;
Lebesgue integral;
Numerical integration;
Gaussian quadrature;
Stokes' theorem;
Green's theorem;
exterior calculus;
De Rham cohomology;
Hodge theory;**



❖ **Antiderivatives**

Optimization Problems

Problem: A farmer has 2400ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Steps In Solving Optimization Problems

- **Understand the Problem** Read the problem carefully. Ask about what is the unknown? What are the given quantities? What are the given conditions?
- **Draw a Diagram** Draw a diagram. Identify the given quantities and required quantities on it.
- **Introduce Notation** Assign a symbol to the quantity that is to be maximized or minimized (let's call it Q for now). Also select symbols ($a, b, c, \dots, x, y$) for other unknown quantities and label the diagram with these symbols (Use initials as suggestive symbols, for example, A for area, h for height, t for time).

Optimization Problems

$$Q(x, t, y)$$

- Express Q in terms of some of the other symbols. If Q has been expressed as a function of more than one variable, use the given information to find relationships (in the form of equations) among these variables. Then use these equations to eliminate all but one of the variables in the expression for Q .
- Q is then expressed as a function of one variable x , say, $Q = f(x)$. Write the domain of this function in the given context.
- Find the absolute maximum or minimum value of $f(x)$. In particular, if the domain of $f(x)$ is a closed interval, then the Closed Interval Method can be used.

Slide 17 of
Lecture 9

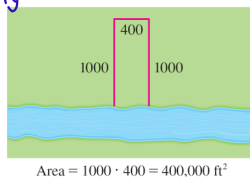
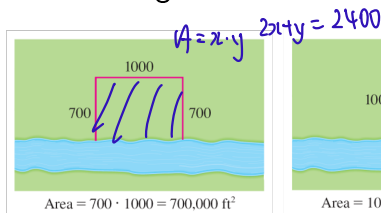
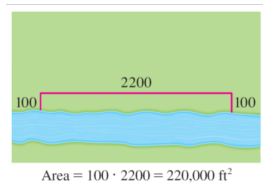
Optimization Problems

Example

A farmer has 2400ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Solution

In order to get a feeling for what is happening in this problem, let's experiment with some specific cases. Figure below shows three possible ways of laying out the 2400ft of fencing.



Optimization Problems

Let x and y be the depth and width of the rectangle (in feet). Then the area A can be expressed in terms of x and y :

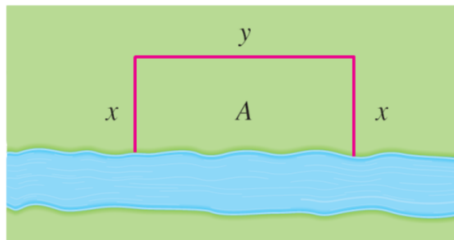
$$A = xy$$

We hope to express A as a function of just one variable, so we eliminate y by expressing it in terms of x . We have

$$2x + y = 2400 \rightarrow y = 2400 - 2x$$

Thus we can have the expression for the area A as following

$$A = xy = x(2400 - 2x) = 2400x - 2x^2$$



Optimization Problems

Note that x can't be negative and the largest x can be is 1200 (why?), so the function to maximize is

polynomial, cont. and diff.

$$A = 2400x - 2x^2 \quad 0 \leq x \leq 1200$$

*$2x + y = 2400$
 $x \geq 0$
 $y \geq 0$*

the critical number is

$$A'(x) = 2400 - 4x = 0 \rightarrow x = 600$$

The maximum value of A must occur either at this critical number or at an endpoint of the interval.

Since $A(0) = 0$, $A(600) = 720000$ and $A(1200) = 0$, the Closed Interval Method gives the maximum value as $A(600) = 720000$.

The corresponding y value is $y = 2400 - 2(600) = 1200$; so the rectangular field should be 600ft deep and 1200ft wide.

Optimization Problems

First Derivative Test for Absolute Extreme Values

Suppose that c is a critical number of a continuous function f defined on an interval.

- (a) If $f'(x) > 0$ for all $x < c$ and $f'(x) < 0$ for all $x > c$, then $f(c)$ is the absolute maximum value of f .
- (b) If $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$, then $f(c)$ is the absolute minimum value of f .

can be closed, open or half close half open

Suppose $\exists a$, $f(a)$ is global maximum. ($a \neq c$)

$$\left\{ \begin{array}{l} a < c, [a, c], (a, c) \Rightarrow \exists x_0 \in (a, c) \frac{f(c) - f(a)}{c - a} = f'(x_0) \xrightarrow{\text{mean value theorem}} \text{contradiction} \\ a > c, [c, a], (c, a) \Rightarrow \exists x_0 \in (c, a) \frac{f(a) - f(c)}{a - c} = f'(x_0) \xrightarrow{\text{contradiction}} \end{array} \right.$$

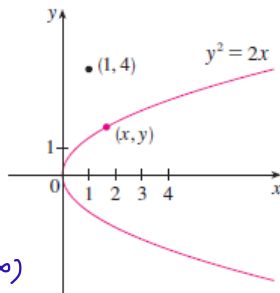
Optimization Problems

Example

Find the point on parabola $y^2 = 2x$ that is closest to the point $(1, 4)$

Solution: The distance between the point $(1, 4)$ and the point (x, y) is

$$y^2 = 2x$$
$$\begin{cases} y = \sqrt{2x} & x \geq 0 \\ y = -\sqrt{2x} & x < 0 \end{cases}$$



$$x \in [0, \infty)$$
$$y \in (-\infty, \infty)$$

$$y^2 = 2x$$
$$x = \frac{y^2}{2}$$

↓ similar to

$$y = \frac{x^2}{2}$$

$$d = \sqrt{(x - 1)^2 + (y - 4)^2}$$

Optimization Problems

But if (x, y) lies on the parabola then $x = \frac{1}{2}y^2$, so the expression for d becomes

$$d = \sqrt{\left(\frac{1}{2}y^2 - 1\right)^2 + (y - 4)^2}$$

Alternatively, we could have substituted $y = \sqrt{2x}$ to get d in terms of x alone. Instead of minimizing d , we minimize its square:

$$f(y) = d^2 = \left(\frac{1}{2}y^2 - 1\right)^2 + (y - 4)^2$$

(The minimum of d occurs at the same point as the minimum of d^2 , but d^2 is easier to work with.)

Differentiating, we obtain

$$f'(y) = 2\left(\frac{1}{2}y^2 - 1\right)y + 2(y - 4) = y^3 - 8$$

so $f'(y) = 0$ when $y = 2$. Observe that $f'(y) < 0$ when $y < 2$ and $f'(y) > 0$ when $y > 2$, so by the First Derivative Test for Absolute Extreme Values, the absolute minimum occurs when $y = 2$.

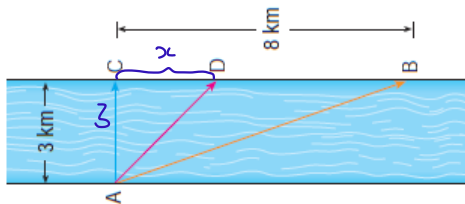
(Or we could simply say that because of the geometric nature of the problem, it is obvious that there is a closest point but not a farthest point.)

The corresponding value of x is $x = \frac{1}{2}y^2 = 2$. Thus the point on $y^2 = 2x$ closest to $(1, 4)$ is $(2, 2)$. [The distance between the points is $d = \sqrt{f(2)} = \sqrt{5}$.]

Optimization Problems

Example

A man launches his boat from point A on a bank of a straight river, 3 km wide, and wants to reach point B , 8 km downstream on the opposite bank, as quickly as possible. He could row his boat directly across the river to point C and then run to B , or he could row directly to B , or he could row to some point D (between C and B) and then run to B . If he can row 6 km/h and run 8 km/h, where should he land to reach as soon as possible? (We assume that the speed of the water is negligible compared with the speed at which the man rows.)



minimise travel time

$$x \in [0, 8]$$

$$|AD| = \sqrt{3^2 + x^2}$$

$$t = \frac{\sqrt{3^2 + x^2}}{6} + \frac{8 - x}{8}$$

Optimization Problems

Solution:

If we let x be the distance from C to D , then the running distance is $|DB| = 8 - x$ and the Pythagorean Theorem gives the rowing distance as $|AD| = \sqrt{x^2 + 9}$. Since we have *time = distance/rate*, the rowing time is $\sqrt{x^2 + 9}/6$ and the running time is $(8 - x)/8$, so the total time T as a function of x is

$$T(x) = \frac{\sqrt{x^2 + 9}}{6} + \frac{8 - x}{8}$$

(river travel) (land travel)

The domain of this function T is $[0, 8]$. Notice that if $x = 0$, he rows to C and if $x = 8$, he rows directly to B . The derivative of T is

$$T'(x) = \frac{x}{6\sqrt{x^2 + 9}} - \frac{1}{8}$$

Thus, using the fact that $x \geq 0$, we have

$$\begin{aligned} T'(x) = 0 &\Leftrightarrow \frac{x}{6\sqrt{x^2 + 9}} = \frac{1}{8} \Leftrightarrow 4x = 3\sqrt{x^2 + 9} \Leftrightarrow 16x^2 \\ &= 9(x^2 + 9) \Leftrightarrow 7x^2 = 81 \Leftrightarrow x = 9/\sqrt{7} \end{aligned}$$

Optimization Problems

The only critical number is $x = \frac{9}{\sqrt{7}}$. To see whether the minimum occurs at this critical number or at an endpoint of the domain $[0, 8]$, we follow the Closed Interval Method by evaluating T at all three points:

$$T(0) = 1.5, \quad T\left(\frac{9}{\sqrt{7}}\right) = 1 + \frac{\sqrt{7}}{8} \approx 1.33, \quad T(8) = \frac{\sqrt{73}}{6} \approx 1.42$$

Since the smallest value occurs when $x = 9/\sqrt{7}$, the absolute minimum value must occur there.

Thus the man should land the boat at a point $9/\sqrt{7} \text{ km}$ ($\approx 3.4 \text{ km}$) downstream from his starting point.

Optimization Problems

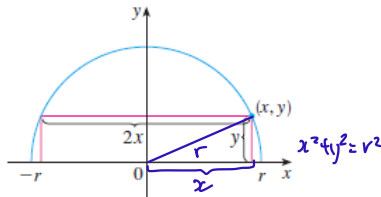
Find the area of the largest rectangle that can be inscribed in a semicircle of radius r .

Solution:

Let's take the semicircle to be the upper half of the circle $x^2 + y^2 = r^2$ with center at the origin. Then the word “inscribed” means that the rectangle has two vertices on the semicircle and two vertices on the x -axis as shown in Figure.

Let (x, y) be the vertex that lies in the first quadrant. Then the rectangle has sides of lengths $2x$ and y , so its area is

$$A = 2xy$$



Optimization Problems

$$A = 2x\sqrt{r^2 - x^2}$$

The domain of the function is $0 \leq x \leq r$. Its derivative is

$$A' = 2\sqrt{r^2 - x^2} - \frac{2x^2}{\sqrt{r^2 - x^2}} = \frac{2(r^2 - 2x^2)}{\sqrt{r^2 - x^2}}$$

which is 0 when $2x^2 = r^2$, that is, $x = \frac{r}{\sqrt{2}}$ (since $x \geq 0$). This value of x gives a maximum value of A since $A(0) = 0$ and $A(r) = 0$. Therefore the area of the largest inscribed rectangle is

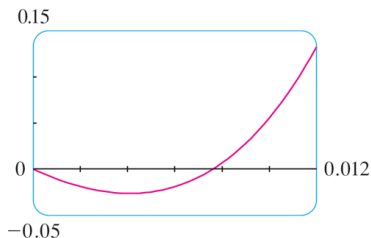
$$A\left(\frac{r}{\sqrt{2}}\right) = 2\frac{r}{\sqrt{2}}\sqrt{r^2 - \frac{r^2}{2}} = r^2$$

Newton's Method

To solve the equation

$$48x(1+x)^{60} - (1+x)^{60} + 1 = 0$$

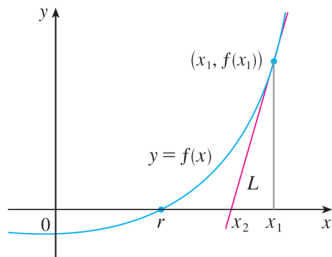
We can find an approximate solution to above equation by using a graphing device.



We see that in addition to the solution $x = 0$, there is a solution between 0.007 and 0.008. Zooming in shows that the root is approximately 0.0076. If we need more accuracy we could zoom in repeatedly.

Newton's Method

Consider the tangent line L to the curve $y = f(x)$ at the point $(x_1, f(x_1))$ and look at the x -intercept of L , labeled x_2 .



The idea behind Newton's method is that the tangent line is close to the curve and so its x -intercept, x_2 , is close to the root r . And x_2 can be easily calculated from tangent line $y - f(x_1) = f'(x_1)(x - x_1)$ by setting $y = 0$,

$$0 - f(x_1) = f'(x_1)(x_2 - x_1) \rightarrow x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \text{if } f'(x_1) \neq 0$$

We use x_2 as a second approximation to r .

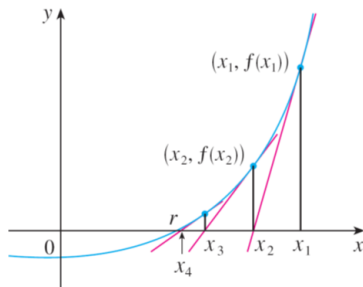
Newton's Method

Next we repeat this procedure with x_1 replaced by the second approximation x_2 , using the tangent line at $(x_2, f(x_2))$.

This gives a third approximation:

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

If we keep repeating this process, we obtain a sequence of approximations $x_1, x_2, x_3, x_4, \dots$ as shown in figure.



Newton's Method

In general, if the n -th approximation is x_n and $f'(x_n) \neq 0$, then the next approximation is given by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

If the numbers x_n become closer and closer to r as n becomes large, then we say that the sequence converges to r and we write

$$\lim_{n \rightarrow \infty} x_n = r \quad \lim_{n \rightarrow \infty} f(x_n) = 0$$

The Newton's method **may fail** if an improper initial approximation x_1 is used. In this case, we need to find another initial value.



Newton's Method

Example

Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

Solution: We apply Newton's method with

$$f(x) = x^3 - 2x - 5 \quad \text{and} \quad f'(x) = 3x^2 - 2$$

Newton used this equation to illustrate his method and he chose $x_1 = 2$ (as $f(1) = -6$, $f(2) = -1$, and $f(3) = 16$).

Newton's formula becomes

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

Newton's Method

with $n = 1$ we have

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = x_1 - \frac{x_1^3 - 2x_1 - 5}{3x_1^2 - 2} = 2 - \frac{2^3 - 2(2) - 5}{3(2)^2 - 2} = 2.1$$

Then with $n = 2$ we obtain

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = x_2 - \frac{x_2^3 - 2x_2 - 5}{3x_2^2 - 2} = 2.1 - \frac{2.1^3 - 2(2.1) - 5}{3(2.1)^2 - 2} = 2.0946$$

It turns out that this third approximation $x_3 \approx 2.0946$ is accurate to four decimal places.

Newton's Method

Example

Use Newton's method to find $\sqrt[6]{2}$ correct to eight decimal places.

Solution:

First we observe that finding $\sqrt[6]{2}$ is equivalent to finding the positive root of the equation

$$x^6 - 2 = 0$$

so we take $f(x) = x^6 - 2$. Then $f'(x) = 6x^5$ and Newton's formula becomes

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^6 - 2}{6x_n^5}$$

Newton's Method

If we choose $x_1 = 1$ as the initial approximation, then we obtain

$$\begin{array}{lcl} x_2 \approx 1.16666667 & \searrow & \Delta x \approx 0.04 \\ x_3 \approx 1.12644368 & \searrow & \Delta x \approx 0.004 \\ x_4 \approx 1.12249707 & \searrow & \Delta x \approx 0.0003 \\ x_5 \approx 1.12246205 & \searrow & \Delta x \approx 0.00000009 \\ x_6 \approx 1.12246205 & \searrow & \Delta x \approx 0.00000000 \end{array}$$

Since x_5 and x_6 agree to eight decimal places, we conclude that

$$\sqrt[6]{2} \approx 1.12246205$$

Newton's Method

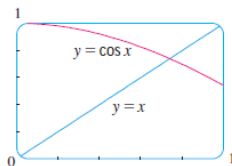
Example

Find, correct to six decimal places, the root of the equation $\cos x = x$.

Solution:

We first rewrite the equation in standard form:

$$\cos x - x = 0$$



Therefore we let $f(x) = \cos x - x$. Then $f'(x) = -\sin x - 1$, so Newton's equation becomes

$$x_{n+1} = x_n - \frac{\cos x_n - x_n}{-\sin x_n - 1} = x_n + \frac{\cos x_n - x_n}{\sin x_n + 1}$$

Newton's Method

In order to guess a suitable value for x_1 , we sketch the graphs of $y = \cos x$ and $y = x$ as shown in figure. It appears that they intersect at a point whose x -coordinate is somewhat less than 1, so let's take $x_1 = 1$ as a convenient first approximation. Then, remembering to put our calculator in radian mode, we get

$$x_2 \approx 0.75036387$$

$$x_3 \approx 0.73911289$$

$$x_4 \approx 0.73908513$$

$$x_5 \approx 0.73908513$$

Since x_4 and x_5 agree to more than six decimal places (eight, in fact), we conclude that the root of the equation, correct to six decimal places, is 0.739085.

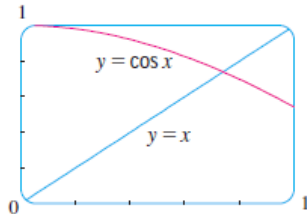
Newton's Method

Instead of using the rough sketch to get a starting approximation for Newton's method in the Example, we could have used the more accurate graph that a calculator or computer provides. Figure below suggests that we use $x_1 = 0.75$ as the initial approximation. Then Newton's method gives

$$x_2 \approx 0.73911114$$

$$x_3 \approx 0.73908513$$

$$x_4 \approx 0.73908513$$



and so we obtain the same answer as before, but with one fewer step.

Antiderivatives

To find a function $F(x)$ whose derivative is a known function $f(x)$. If such a function $F(x)$ exists, it is called an antiderivative of f .

Definition A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

For instance, let $f(x) = x^2$. It is not difficult to find an antiderivative of $f(x)$. In fact, if $F(x) = \frac{1}{3}x^3$, then $F'(x) = x^2 = f(x)$.

But the function $G(x) = \frac{1}{3}x^3 + 100$ also satisfies $G'(x) = x^2$. Therefore both $F(x)$ and $G(x)$ are antiderivatives of $f(x)$.

Indeed, any function of the form $H(x) = \frac{1}{3}x^3 + C$, where C is a constant, is an antiderivative of $f(x)$.

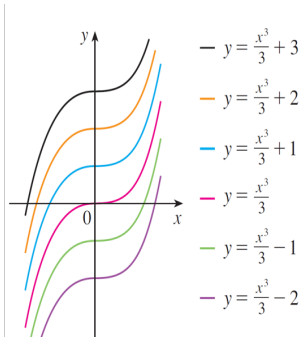
Antiderivatives

1 Theorem If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

By assigning specific values to the constant C , we obtain a family of functions whose graphs are vertical translates of one another.



Antiderivatives

Example

Find the most general antiderivative of each of the following functions.

(a) $f(x) = \sin(x)$ (b) $f(x) = x^n, n \geq 0$ (c) $f(x) = x^{-3}$

Solution:

(a) If $F(x) = -\cos(x)$, then $F'(x) = \sin(x)$, so an antiderivative of $\sin(x)$ is $-\cos(x)$. By Theorem 1, the most general antiderivative is $G(x) = -\cos x + C$.

(b) We use the Power Rule to discover an antiderivative of x^n :

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = \frac{(n+1)x^n}{n+1} = x^n$$

Therefore the general antiderivative of $f(x) = x^n$ is

$$F(x) = \frac{x^{n+1}}{n+1} + C$$

Antiderivatives

(c) If we put $n = -3$ in the antiderivative formula from part(b), we get the particular antiderivative $F(x) = x^{-2}/(-2)$.

But notice that $f(x) = x^{-3}$ is not defined at $x = 0$.

$\Rightarrow (-\infty, 0) \cup (0, \infty)$

Thus Theorem 1 tells us only that the general antiderivative of f is $x^{-2}/(-2) + C$ on any interval that does not contain 0. So the general antiderivative of $f(x) = \frac{1}{x^3}$ is

$$F(x) = \begin{cases} -\frac{1}{2x^2} + C_1 & \text{if } \underline{x > 0} \\ -\frac{1}{2x^2} + C_2 & \text{if } \underline{x < 0} \end{cases} \quad \text{interval}$$

Antiderivatives

2 Table of Antidifferentiation Formulas

$$\begin{cases} F'(x) = f(x) \\ G'(x) = g(x) \end{cases}$$

$$f(x) - g(x) \Rightarrow F(x) - G(x)$$

$$f(x) \cdot g(x), \frac{f(x)}{g(x)} \quad \times$$

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\cos x$	$\sin x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sin x$	$-\cos x$
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\sec^2 x$	$\tan x$
		$\sec x \tan x$	$\sec x$

Each formula in the table is true because the derivative of the function in the right column appears in the left column.

In particular, the first formula says that the antiderivative of a constant times a function is the constant times the antiderivative of the function.

The second formula says that the antiderivative of a sum is the sum of the antiderivatives. (We use the notation $F' = f$, $G' = g$.)

Antiderivatives

Example

Find all the functions g such that,

$$g'(x) = 4 \sin x + \frac{2x^5 - \sqrt{x}}{x}$$

No quotient value

Solution: We first rewrite the given function as follows:

$$g'(x) = 4 \sin x + \frac{2x^5}{x} - \frac{\sqrt{x}}{x} = 4 \sin x + 2x^4 - \frac{1}{\sqrt{x}}$$

Thus we want the antiderivative of

$$g'(x) = 4 \sin x + 2x^4 - x^{-1/2}$$

Using the formulas in Table 2 together with Theorem 1, we obtain

$$g(x) = 4(-\cos x) + 2\frac{x^5}{5} - \frac{x^{1/2}}{\frac{1}{2}} + C = -4\cos x + \frac{2}{5}x^5 - 2\sqrt{x} + C$$

Antiderivatives

Example

Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, and $f(1) = 1$

Solution:

The general antiderivative of $f''(x) = 12x^2 + 6x - 4$

$$f'(x) = 12\frac{x^3}{3} + 6\frac{x^2}{2} - 4x + C = 4x^3 + 3x^2 - 4x + C$$

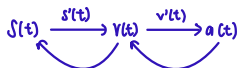
Using the antidifferentiation rules once more, we find that

$$f(x) = 4\frac{x^4}{4} + 3\frac{x^3}{3} - 4\frac{x^2}{2} + Cx + D = x^4 + x^3 - 2x^2 + Cx + D$$

To determine C and D , we use the given conditions that $f(0) = 4$ and $f(1) = 1$. From $f(0) = 0 + D = 4$, we have $D = 4$. From $f(1) = 1 + 1 - 2 + C + 4 = 1$, we have $C = -3$. Therefore the required function is

$$f(x) = x^4 + x^3 - 2x^2 - 3x + 4$$

Antiderivatives



Antidifferentiation is particularly useful in analyzing the motion of an object moving in a straight line. We know if the object has position function $s(t)$, then the velocity function is $v(t) = s'(t)$.

This means that the position function is an antiderivative of the velocity function.

Likewise, the acceleration function is $a(t) = v'(t)$, so the velocity function is an antiderivative of the acceleration.

If the acceleration and the initial values $s(0)$ and $v(0)$ are known, then the position function can be found by antidifferentiation twice

Antiderivatives

Example

A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6\text{cm/s}$ and its initial displacement is $s(0) = 9\text{cm}$. Find its position function $s(t)$.

Solution:

Since $v'(t) = a(t) = 6t + 4$, antidifferentiation gives

$$v(t) = 6\frac{t^2}{2} + 4t + C = 3t^2 + 4t + C$$

From $v(0) = C$ and $v(0) = -6\text{cm/s}$, we have $v(t) = 3t^2 + 4t - 6$. Since $v(t) = s'(t)$, $s(t)$ is the antiderivative of $v(t)$, we have

$$s(t) = 3\frac{t^3}{3} + 4\frac{t^2}{2} - 6t + D = t^3 + 2t^2 - 6t + D$$

From $s(0) = D$ and $s(0) = 9\text{cm}$, we have $D = 9\text{cm}$ and $s(t)$ is

$$s(t) = t^3 + 2t^2 - 6t + 9$$

Antiderivatives

Example

A ball is thrown upward with a speed of 15m/s from the edge of a cliff 140m above the ground. Find its height above the ground t seconds later. When does it reach its maximum height? When does it hit the ground? (The acceleration on earth surface is m/s^2)

Solution:

The ball moves upward, so upward is positive direction. The distance above the ground is $s(t)$, and the acceleration $a(t)$ is,

$$a(t) = \frac{dv}{dt} = -9.8$$

Taking antiderivatives, we have the velocity $v(t)$,

$$v(t) = -9.8t + C$$

We have $v(0) = 15$, so $15 = 0 + C$, that is $C = 15$. Further

$$v(t) = -9.8t + 15$$

Antiderivatives

The maximum height is reached when $v(t) = 0$, that is, after $15/9.8 \approx 1.53s$. Since $s'(t) = v(t)$, we use antidifferentiation again and obtain

$$s(t) = -4.9t^2 + 15t + D$$

Using the condition $s(0) = 140$, we have $140 = 0 + D$, that is $D = 140$, and

$$s(t) = -4.9t^2 + 15t + 140$$

The expression for $s(t)$ is valid until the ball hits the ground. This happens when $s(t) = 0$; that is, when

$$4.9t^2 - 15t - 140 = 0$$

Using the quadratic formula to solve this equation, we get

$$t = \frac{15 \pm \sqrt{2969}}{9.8} \approx 7.1s \quad (\text{Ignore negative value})$$

Understanding Checker

Q1. If $f'(x) = (x-2)(x-3)(x-4)^3$, ^{→ function is diff. every where} and local maximal and minimal can be found at $x=a$ and $b=a$ respectively. Then we have

(A) $a=2, 3; b=4$ (B) $a=2; b=3, 4$ (C) $a=4; b=2$ (D) $a=2; b=2$

Let $f'(x) = 0$

$\therefore x=2, x=3, x=4 \rightarrow$ critical numbers

$(-\infty, 2) \cup (2, 3) \cup (3, 4) \cup (4, \infty)$

-	+	+	+	$x-2$
+	+	+	+	$(x-3)^2$
-	-	-	+	$(x-4)^3$
+	-	-	+	$f'(x)$

$\therefore D$

Q2. What is the horizontal asymptote for $f(x) = \sqrt{x^2 + 4x + 3} - x$

(A) 2 (B) $x=2$ (C) $y=2$ (D) D.N.E

* horizontal asymptote is $y = \dots$

$\lim_{x \rightarrow \infty / -\infty} f(x) = L, y=L \rightarrow$ horizontal asymptote

$$\begin{aligned} & \lim_{x \rightarrow \infty} \sqrt{x^2 + 4x + 3} - x \\ &= \lim_{x \rightarrow \infty} \frac{4x + 3}{\sqrt{x^2 + 4x + 3} + x} \quad \text{looking @ } \sqrt{x^2 + 4x + 3} - x \\ &= \lim_{x \rightarrow \infty} \frac{4 + \frac{3}{x}}{\sqrt{1 + \frac{4}{x} + \frac{3}{x^2}} + 1} = 2 \quad \therefore \lim_{x \rightarrow \infty} = \infty \end{aligned}$$

*Handwritten notes: $\rightarrow \lim_{x \rightarrow -\infty} \frac{4 + 3/x}{-\sqrt{1 + 4/x + 3/x^2} + 1} \rightarrow$ becomes 0.
 $\uparrow \infty$ $\downarrow +\infty$*